

NATHAN ZHANG

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RESEARCH EXPERIENCE

Research Assistant, UBC Sauder School of Business (Apr 2026 – Present)

Supervisors: J. Frank Li and Cheng Chen

- As the sole RA on a faculty-led study of age-biased technological change, independently compare empirical measures and panel fixed-effects regression designs for wage and occupational sorting models. Link the CPS labor survey, O*NET occupation database, and USPTO patent data.
- Harmonized 60+ O*NET releases. Use PatentsBERTa and Qwen to generate embeddings and extract features from task and patent text, comparing measures of technological change.
- Implemented GMM, block bootstrap inference, calibration, and robustness checks. Drafted the data section. Maintain the R/Python replication package and write Bash scripts and Slurm job scripts for HPC workloads.

Research Assistant, UBC Sauder School of Business (May 2026 – Present)

Supervisor: Allen Hu

- Evaluated financial-news coverage and reliability by matching RavenPack metadata to over 3.2 million ProQuest articles, checking missingness and publication-level coverage.
- Built a reproducible creator-video panel pipeline for financial-media research. Linked multimodal assets and metadata with source tracking and validation to support auditable sample construction.

Research Assistant, Vancouver School of Economics at UBC (Jan 2025 – May 2026)

Supervisors: Jonathan Graves and Laura Nelson

- Developed Python/R data science and machine learning materials for economics and sociology courses through COMET and Praxis UBC, applying NLP, computer vision, and time-series forecasting. The platforms have served 1,000+ cumulative student users.
- Co-developed a natural language inference (NLI) tutorial on zero-shot stance classification and human-label validation, in first revision at Programming Historian. Designed and taught a computer vision workshop.

PUBLICATIONS AND MANUSCRIPTS

Memory Is the Difference between Two Futures: Memory Response Paths and Path-Structured Context Compilation (2026)

Manuscript submitted, under review. Coauthored with Jerrold Huang

Defines a memory's effect as the difference between matched futures with and without it, traced over time. In a separate 144-item evaluation, PSCC's LLM-judged quality is 6.94 percentage points higher than A-Mem's (95% CI: 2.64 to 11.32) while using 67.2% of its API tokens.

When Answers Surprise (2026)

Accepted at ICAIF 2026. Coauthored with Jerrold Huang

Tests whether the novelty of a management answer predicts returns beyond earnings surprise. Built an earnings-call panel linking Capital IQ transcripts, IBES, CRSP, and historical S&P 500 membership across 2018-2025. Evaluation uses time holdouts, learner checks, placebos, and timing checks.

EDUCATION

The University of British Columbia (Sep 2023 – May 2028)

B.A., Combined Major in Economics and Statistics, Minor in Data Science, Vancouver, BC

- Cumulative average: 93.5/100

RESEARCH INTERESTS

Methods: Statistical learning and inference, causal inference and econometrics, panel and time-series analysis, text and image methods.

Applications: Technological change and labor, financial risk and ESG, computational social science.

SELECTED PROJECTS

Double Machine Learning for Payment-Friction Spending Effects (2025)

Course research project [Code] [Report]

- Used cross-fitted DoubleML on about 5,600 respondents and roughly 22,000 in-person person-days. Under selection on observables and overlap, the estimated conditional effect of lower-friction payment mixes on daily spending is about 5%, with heterogeneity by ticket size, income, and education.

Canadian Inflation Forecasting and Macro Signal Evaluation (2025)

Forecasting project [Code]

- Built an R/Quarto VAR forecast for Canadian inflation with rolling out-of-sample tests against naive and random-walk baselines. Presented impulse responses in a Bank of Canada-style policy briefing.

TEACHING AND RESEARCH INFRASTRUCTURE

Image Analysis Workshop: Computer Vision for Social Science Research (Mar 2026)

Workshop lead and instructor [Workshop]

Designed, hosted, and taught a Praxis UBC workshop that moved from preprocessing and BoVW to CNN/ViT embeddings, classification, object detection, OCR, and captioning.

A Study of Richter's Kouroi Through Image Embedding (Jul 2025)

Coauthor with Yash Mali and Krishaant Pathman [Notebook]

Used ConvNeXt V2 embeddings, PCA, clustering, and classification to test whether image embeddings recover structure in Richter's kouroi categories.

LLM Prediction Module (2025)

Coauthor [Module]

Coauthored course material on next-word prediction, token probabilities, and economics applications.

PROFESSIONAL EXPERIENCE

ESG Research Analyst Co-op, Capital Markets & Credit Investments (Sep 2026 – Present)

British Columbia Investment Management Corporation (BCI), Vancouver, BC

- Conduct quantitative analysis to discover governance risk factors, focusing on executive compensation, firm risk, and returns. Use Databricks and Claude Code to backtest simulated governance strategies and assess shareholder interest alignment, supporting fundamental investment research, BCI's proxy voting strategy, and policy updates.

Teaching Assistant (Sep 2024 – Aug 2025)

UBC Chinese Language Program, Vancouver, BC

- Supported Chinese language teaching, grading, volunteer training, and feedback coordination.

AWARDS AND SCHOLARSHIPS

Work Learn International Undergraduate Research Awards (WLIURA) (UBC), 2026

Economics Undergraduate Scholarship (Vancouver School of Economics, UBC), 2025

Faculty of Arts International Student Scholarship (UBC Faculty of Arts), 2024, 2025

SELECTED COURSEWORK

Statistical Inference (100), Machine Learning and Data Mining (93), Causal Methods (98)

Advanced Econometrics (98), Statistical Learning (97), Time Series and Forecasting (95)

TECHNICAL SKILLS AND LANGUAGES

Programming & data: Python, R, SQL, Bash, Julia, PySpark, DuckDB, Polars, BQL

Methods: econometrics, factor models, causal inference, panel/time series, machine learning, NLP, computer vision

Modeling libraries: NumPy, SciPy, scikit-learn, PyTorch, statsmodels, fixest, OpenCV, spaCy, NLTK

Platforms & tools: Databricks, Bloomberg, Claude Code, DevOps/Git, GitHub Actions, HPC/Slurm, Jupyter/Quarto

Languages: Mandarin Chinese: native / English: academic working proficiency / Japanese: JLPT N3